



Figure 1: A simple window

you prefer, instead of *kate*). For the OpenGL headers, run `'sudo apt-get install freeglut3-dev'`.

The 'Hello World' first step

Now for the classic first-time program. Open the editor, and enter the following code:

```
#include<GL/glut.h>

void main(int argc, char**argv) {
    glutInit(&argc, argv);
    glutInitWindowPosition(100,100);
    glutInitWindowSize(500,500);
    glutCreateWindow("Hello World");
    glutMainLoop();
}
```

Save the file as `'basic.c'`, and to compile it, run `'gcc basic.c -l glut'` at the terminal. Run the compiled binary with `./a.out`. Here, we have imported `glut.h` (in the *GL* folder). To locate it, run `'whereis GL'` and it will give you the installed location, like *GL*: `/usr/include/GL`. There are just five basic functions used to create a window for drawing. The first, `glutInit()`, initialises the display. Then `glutInitWindowPosition(int x, int y)` specifies the screen location (upper-left corner) of the window. Next, `glutInitWindowSize(int x, int y)` specifies the size of the window; `glutCreateWindow(char *string)` names the window for identification, and creates it; and `glutMainLoop()` is used to display the window and begin event processing. Figure 1 shows you how it appears.

The compilation command specified `-l glut`, which means, a link with the glut library file. By default, GCC names the output compiled binary `'a.out'`; to change it, specify the `-o` (output file name) switch—for example, `'gcc begin.c -l glut -o hello'`.

Change the background colour of a window

The OpenGL function `glutDisplayFunc(void function_name)` is used to specify a user function that will handle the display of a window. Let's use this to change the colour of a window:



Figure 2: The window after adding colours

```
#include<GL/glut.h>
#include<GL/gl.h>
void display() {
    glClearColor(1,0,0,0);
    glClear(GL_COLOR_BUFFER_BIT);
    glFlush();
}

void main(int argc, char**argv) {
    glutInit(&argc, argv);
    glutInitWindowPosition(100,100);
    glutInitWindowSize(500,500);
    glutCreateWindow("Hello World");
    glutDisplayFunc(display);
    glutMainLoop();
}
```

Save it as `'basic2.c'`, compile and run it as in the earlier example. The result is shown in Figure 2.

The three GL functions we used to colour the window background are:

- `glClearColor(R, G, B, alpha)`: This is used to set the colour for a window. The numbers you can use for each colour are between 0 and 1; you can use float numbers like 0.1, 0.11, etc.
- `glClear(GL_COLOR_BUFFER_BIT)`: This clears the screen to the desired colour set by `glClearColor`.
- `glFlush()` executes the commands to the screen rather than storing it in a buffer.

In the next part of the series, I plan to explore how to draw lines and triangles, and use keyboard inputs. 

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By: Rajnish Kumar

The author has spent his free time learning open source technologies. His areas of interest are GCC, Linux, courses, Python, SciLab, PHP, MySQL, Wordpress, OpenGL, OpenCV, Arduino, etc. You can reach him at rajnishs91@gmail.com.